Answers & Solutions:

1. **C. Resistor noise**

2. **B. The square wave has an infinite number of harmonics, whereas the sinewave has only one.**

3. **D. All of the above**

4. **A. Input noise voltage**

5. **C. pink noise**

6. **A. a shift in the input signal also results in the corresponding shift in the output**

7. **C. atmospheric noise**

8. **A. 0.006 µA**

   Solution:

$$I_n = \sqrt{2q_e I_{dc} BW}$$
$$I_n = \sqrt{2\left(1.6x10^{-19}\right)\left(1.40\text{ mA}\right)\left(80x10^{3}\right)}$$
$$I_n = 6\text{ nA} = 0.006\ \mu\text{A} \rightarrow \text{Ans.}$$

9. **C. sinusoid signals multiplied by decaying exponentials**

10. **D. All of the above**

11. **B. estimating the area under the square of its amplitude spectrum**

12. **B. signal power divided by noise power**

13. **C. Fourier Transform**

14. **B. its temperature**

15. **A. a shift in the input signal also results in the corresponding shift in the output**

16. **C. equal power per octave**

17. **C. Fourier Transform**

18. **B. greater at low frequencies**

19. **A. Parseval's Theorem**

20. **B. in the channel**

21. **B. Complementary error function goes on decreasing**

22. **D. None of the above**

23. **A. transistors and diodes**

24. **D. 398 mA**

   Solution:

$$I_n = \sqrt{2q_e\left(2I_o + I_{dc}\right)BW}$$
$$I_{dc} = \frac{I_n{}^2}{2q_e BW}$$
$$I_{dc} = \frac{\left(0.16x10^{-6}\right)^2}{2\left(1.609x10^{-19}\right)\left(200x10^{3}\right)}$$
$$I_{dc} = 0.398\text{ A} = 398\text{ mA} \rightarrow \text{Ans.}$$

where:

$$I_n = \frac{V_n}{R} = \frac{12\ \mu\text{V}}{75\ \Omega} = 0.16\ \mu\text{A}$$

25. **C. how much noise an amplifier adds to a signal**

END